Fiscal Year:	FY 2018	**Task Last Updated:**	FY 03/20/2018
PI Name:	Mudawar, Issam Ph.D.		
Project Title:	Flow Boiling and Condensation Experiment (FBCE)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mudawar@ecn.purdue.edu	**Fax:**	FY 765-494-0539
PI Organization Type:	UNIVERSITY	**Phone:**	765-494-5705
Organization Name:	Purdue University		
PI Address 1:	Mechanical Engineering Building		
PI Address 2:	585 Purdue Mall		
PI Web Page:	https://		
City:	West Lafayette	**State:**	IN
Zip Code:	47907-1288	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2017	**End Date:**	12/31/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
Contact Email:	david.f.chao@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hasan, Mohammad Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	NNX17AK98G		
Performance Goal No.:			
Performance Goal Text:			
	EDITOR'S NOTE: Continuation of "Flow Boiling and Condensation Experiment (FBCE)," grants NNX13AB01G and NNX12AK14G, which covered from 6/1/2012-5/31/2017. Several efforts are presently underway to reduce the size and weight of space systems by transitioning from single-phase to two-phase thermal management. Because of large density differences between liquid and vapor, buoyancy can play an important role in defining the motion of vapor relative to liquid and, therefore, influence heat transfer effectiveness. This is particularly the case for future space missions, including the manned mission to Mars. The emphasis in two-phase fluid physics is evident from several recent NASA workshops that culminated in critical recommendations concerning Rankine cycle power conversion, thermal control systems, and advanced life support systems. Among others, key fluid physics mechanisms that have been identified are flow boiling and condensation. A key strategy is to develop mechanistic models for flow conditions that would ensure insensitivity to gravity level for these important heat transfer		

Task Description:	mechanisms. This would allow performing most of the required confirmatory experiments on Earth. Those topics and the minimum flow requirements constitute key objectives for the proposed study. Unfortunately, most of the two-phase heat transfer know-how amassed over nearly a century of research comes from experiments conducted in Earth's gravity. The proposed study aims to develop an experimentally validated, mechanistic model for the minimum flow criteria to ensure gravity independent flow boiling heat transfer and condensation. Parabolic flight experiments and preparations for International Space Station (ISS) experiments will occur throughout the course of the proposed study. This project will be a joint effort between the Purdue University Boiling and Two-Phase Flow Laboratory (PU-BTPFL) and the NASA Glenn Research Center. Personnel from the two organizations combine extensive experience in research and development of both flow boiling and condensation systems, and in conducting microgravity experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed research will lay the foundation of development of advanced high power to weight ratio thermal management systems for NASA's future space vehicles and planetary bases.
Task Progress:	Modeling of two-phase flow transient behavior and instabilities has traditionally been one of the more challenging endeavors in heat transfer research due to the need to distinguish between a wide range of instability modes systems can manifest depending on differences in operating conditions, as well as the difficulty in experimentally determining key characteristics of these phenomena. This study presents a new mechanistic model for Density Wave Oscillations (DWOs) in vertical upflow boiling using conclusions drawn from analysis of flow visualization images and transient experimental results as a basis from which to begin modeling. Counter to many prior studies attributing DWOs to feedback effects between flow rate, pressure drop, and flow enthalpy causing oscillations in position of the bulk boiling boundary, the present instability mode stems primarily from body force acting on liquid and vapor phases in a separated flow regime leading to liquid accumulation in the near-inlet region of the test section, which eventually departs and moves along the channel, acting to re-wet liquid film along the channel walls and re-establish annular, co-current flow. This process was modeled by dividing the test section into three distinct control volumes and solving transient conservation equations for each, yielding predictions of frequencies at which this process occurs as well as amplitude of associated pressure oscillations. Values for these parameters were validated against an experimental database of 236 FC-72 points and show the model provides good predictive accuracy and capably captures the influence of parametric changes to operating conditions.
Bibliography Type:	Description: (Last Updated: 01/06/2026)
Abstracts for Journals and Proceedings	Mudawar I, O'Neill LE, Lee S, Devahdhanush VS, Ganesan V, Lee J, Hasan MM, Nahra HK, Balasubramaniam R, Mackey JR. "Flow Boiling and Condensation Experiment (FBCE) for the International Space Station." 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Articles in Peer-reviewed Journals	Kharangate CR, Mudawar I. "Review of computational studies on boiling and condensation." International Journal of Heat and Mass Transfer. 2017 May;108 Pt A;1164-96. https://doi.org/10.1016/j.ijheatmasstransfer.2016.12.065 , May-2017
Articles in Peer-reviewed Journals	Mudawar I. "Flow boiling and flow condensation in reduced gravity." Advances in Heat Transfer. 2017;49:225-306. https://doi.org/10.1016/bs.aiht.2017.06.002 , Aug-2017
Articles in Peer-reviewed Journals	Lee S, Devahdhanush VS, Mudawar I. "Pressure drop characteristics of large length-to-diameter two-phase micro-channel heat sinks." International Journal of Heat and Mass Transfer. 2017 Dec;115 Pt B:1258-75. https://doi.org/10.1016/j.ijheatmasstransfer.2017.08.104 , Dec-2017
Articles in Peer-reviewed Journals	Lee S, Devahdhanush VS, Mudawar I. "Frequency analysis of pressure oscillations in large length-to-diameter two-phase micro-channel heat sinks." International Journal of Heat and Mass Transfer. 2018 Jan;116:273-91. https://doi.org/10.1016/j.ijheatmasstransfer.2017.08.107 , Jan-2018
Articles in Peer-reviewed Journals	O'Neill LE, Mudawar I, Hasan MM, Nahra HK, Ramaswamy B, Hall NR, Lokey A, Mackey JR. "Experimental investigation into the impact of density wave oscillations on flow boiling system dynamic behavior and stability." International Journal of Heat and Mass Transfer. 2018 May;120:144-66. Available online 18 December 2017. https://doi.org/10.1016/j.ijheatmasstransfer.2017.12.011 , May-2018